

NANYANG TECHNOLOGICAL UNIVERSITY**SEMESTER 2 EXAMINATION 2024-2025****EE6427 – VIDEO SIGNAL PROCESSING**

April / May 2025

Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 4 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

1. (a) In a compression scheme, a data source consists of eight symbols, with the probability distribution is given in Table 1.

Table 1

Symbol	S ₀	S ₁	S ₂	S ₃	S ₄	S ₅	S ₆	S ₇
Probability of occurrence	0.03	0.05	0.07	0.11	0.14	0.16	0.20	0.24

- (i) Design a suitable set of Huffman codewords for the eight symbols. Clearly show all the key steps and calculations.
- (ii) Consider a data stream X consisting of 4000 symbols with probability distribution given in Table 1. The data stream is to be compressed using the Huffman codewords designed in part (i). Find the expected (average) storage requirement of data stream X in bytes.

Note: Question No. 1 continues on page 2.

- (iii) A student would like to compress a particular data sequence consisting of 10 symbols. Each symbol in the sequence can be any of the eight symbols given in Table 1. The student uses two different schemes to represent the sequence.

Uncompressed scheme: 4 bits is used to represent each symbol.

Compressed scheme: Huffman codewords designed in part (i) are used to compress the symbols.

Find the lowest possible compression ratio that can be obtained for the sequence.

(18 Marks)

- (b) Briefly explain how Huffman coding is used in the P-frame encoding of the MPEG-1 standard.

(7 Marks)

2. (a) In a motion estimation and compensation scheme for video compression, a 4×4 pixel block is shown in Figure 1 and its co-located block in the reference frame is shown by the bounding box in Figure 2. The search window is within ± 1 pixels. The metric used for motion estimation is sum of absolute errors.

				0	0	0	0	0	0
10	10	10	10	10	10	10	10	10	10
20	20	20	20	10	10	10	10	10	10
30	30	30	30	18	20	20	20	20	15
40	40	40	40	34	30	30	30	32	26
				36	40	40	38	40	42

Figure 1

Figure 2

- (i) Find the motion vector (M_u, M_v) within the search window, where M_u is the row displacement (positive direction is pointing downwards) and M_v is the column displacement (positive direction is pointing to the right).
- (ii) Find the best matched block in the reference frame and write down the prediction error $E(i, j)$.

Note: Question No. 2 continues on page 3.

- (iii) The prediction error $E(i, j)$ undergoes the Integer Transform with transform matrix given by:

$$H = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & -1 & -2 \\ 1 & -1 & -1 & 1 \\ 1 & -2 & 2 & -1 \end{bmatrix}.$$

Find the Integer Transform of the prediction error $E(i, j)$.

(17 Marks)

- (b) Briefly explain the effect(s) of increasing the Quantization Parameter (QP) on the video bitrate and video quality in the H.264 video compression.

(8 Marks)

3. (a) A Long Short-Term Memory (LSTM) network has the following settings:

Initial hidden state, $\mathbf{h}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, Initial cell state, $\mathbf{c}_0 = \begin{bmatrix} 0.2 \\ 0.4 \end{bmatrix}$,

Input gate weight matrix, $\mathbf{W}_i = [\mathbf{W}_{hi} \quad \mathbf{W}_{xi}] = \begin{bmatrix} 0.4 & 0.3 & 0.5 & 0.4 \\ 0.2 & 0.1 & 0.3 & 0.2 \end{bmatrix}$,

Gate gate (also known as candidate gate) at timestep $t=1$, $\mathbf{g}_1 = \begin{bmatrix} 0.2 \\ 0.3 \end{bmatrix}$,

Forget gate at timestep $t=1$, $\mathbf{f}_1 = \begin{bmatrix} 0.8 \\ 0.9 \end{bmatrix}$,

Output gate at timestep $t=1$, $\mathbf{o}_1 = \begin{bmatrix} 0.7 \\ 0.8 \end{bmatrix}$,

A 2-timestep input is given by $\mathbf{x} = [\mathbf{x}_1 \quad \mathbf{x}_2]$ where $\mathbf{x}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Assume no bias is used in the computation of the LSTM. The sigmoid and tanh functions are given as follows.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Note: Question No. 3 continues on page 4.

- (i) Find the Input gate $\mathbf{i}_1$ at timestep $t = 1$.
- (ii) Find the cell state $\mathbf{c}_1$ at timestep $t = 1$.
- (iii) Find the Input gate $\mathbf{i}_2$ at timestep $t = 2$. Round your result to 3 decimal places.

(17 Marks)

- (b) A user would like to develop a video clip content summarization application that takes in a video clip and generates a text sentence to describe the video content. The user is planning to use LSTM(s) to achieve the goal. The user has extracted the feature embedding for each frame of the video clips. Briefly list two main steps of how to use LSTM(s) to address this application.

(8 Marks)

4. (a) Draw a diagram of the general architecture of a two-stage object detector. Clearly label all the key components in the diagram.

(8 Marks)

- (b) Draw a diagram of the Transformer Decoder structure in the Transformer model. Your diagram should include the output embedding and all other key components in the Transformer Decoder.

(9 Marks)

- (c) With the aid of a simple diagram, briefly explain how parallel adaptor can be used to finetune a Transformer-based Foundation Model.

(8 Marks)

END OF PAPER

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1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.